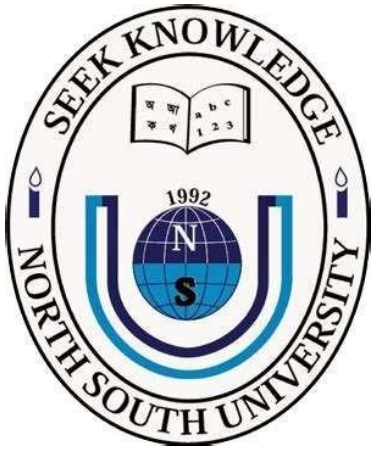
Analog Electronics II: Op-amp & 555 Timer



Course instructor:
Dr. Mohsin Sajjad
Assistant Professor
North South University

EEE 311

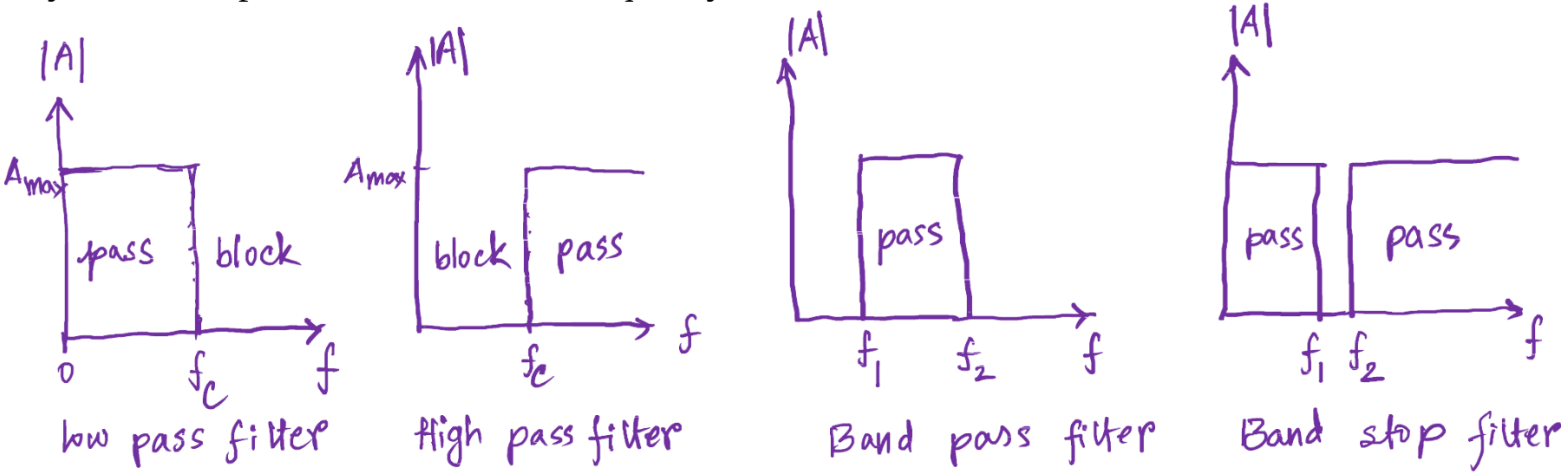
Lecture # 9

Frequency

The **frequency response** of a circuit is the variation in its behavior with change in signal frequency.

The sinusoidal steady-state frequency responses of circuits are of significance in many applications, especially in communications and control systems.

A specific application is in electric filters that block out or eliminate signals with unwanted frequencies and pass signals of the desired frequencies. Filters are used in radio, TV, and telephone systems to separate one broadcast frequency from another.



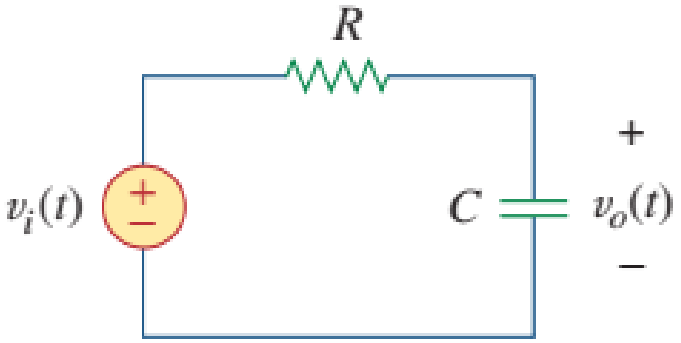
Source: Alexander-Sadiku Chapter 14

Active Filters vs Passive Filters

Basis for Comparison	Active Filter	Passive Filter
Composed of	Active components like op-amp, transistor etc.	Passive components like resistor, inductor and capacitor etc.
Cost	High	Comparatively low.
Circuit complexity	More complex	Less complex than active filter.
Weight	Low	Comparatively bulkier due to presence of inductors.
Q factor	High	Very low in comparison to active filters.
External power supply	Required	Not required
Sensitivity	More sensitive	Comparatively less sensitive.

Passive Filters: Lowpass Filter

A typical lowpass filter is formed when the output of an *RC* circuit taken off the capacitor



The transfer function $$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{1/j\omega C}{R + 1/j\omega C}$$

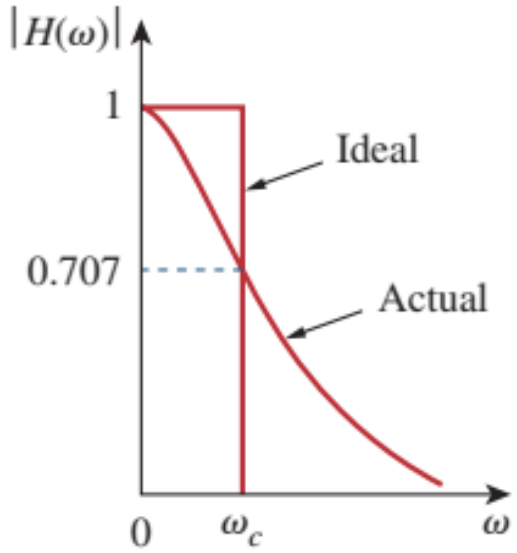
$$\mathbf{H}(\omega) = \frac{1}{1 + j\omega RC}$$

Note that $\mathbf{H}(0) = 1, \mathbf{H}(\infty) = 0.$

the *cutoff frequency* ω_c, is obtained by setting the magnitude of $\mathbf{H}(\omega)$ equal to $1/\sqrt{2}$,

$$H(\omega_c) = \frac{1}{\sqrt{1 + \omega_c^2 R^2 C^2}} = \frac{1}{\sqrt{2}}$$

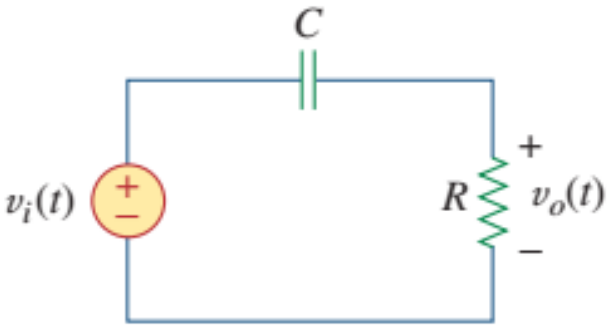
$$\omega_c = \frac{1}{RC}$$



A **lowpass filter** is designed to pass only frequencies from dc up to the cutoff frequency ω_c.

Passive Filters: Highpass Filter

A highpass filter is formed when the output of an *RC* circuit is taken off the resistor

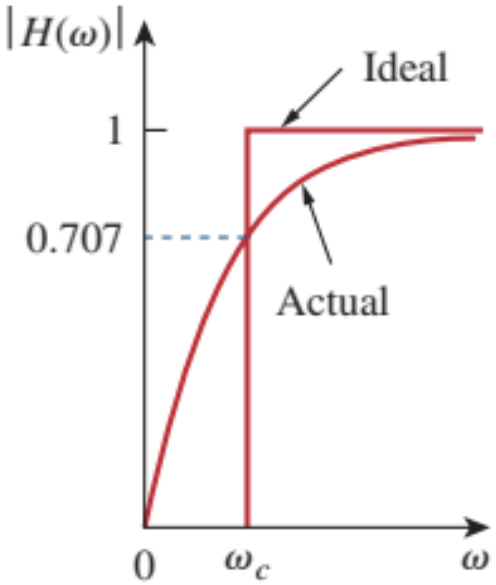


The transfer function is

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{R}{R + 1/j\omega C}$$

$$\mathbf{H}(\omega) = \frac{j\omega RC}{1 + j\omega RC}$$

Note that $\mathbf{H}(0) = 0$, $\mathbf{H}(\infty) = 1$



the corner or cutoff frequency is

$$\omega_c = \frac{1}{RC}$$

A highpass filter is designed to pass all frequencies above its cutoff frequency ω_c.

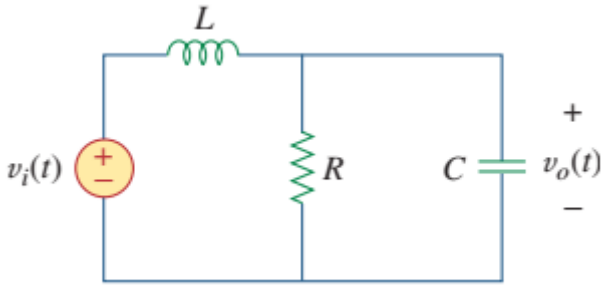
Passive Filters:

A lowpass filter can also be formed when the output of an *RL* circuit is taken off the resistor. Of course, there are many other circuits for lowpass filters.

A highpass filter can also be formed when the output of an *RL* circuit is taken off the inductor.

Example Problem 14.10

Determine what type of filter is shown in Fig. 14.39. Calculate the corner or cutoff frequency. Take $R = 2\ \text{k}\Omega$, $L = 2\ \text{H}$, and $C = 2\ \mu\text{F}$.



Solution: The transfer function is

$$\mathbf{H}(s) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{R \parallel 1/sC}{sL + R \parallel 1/sC}, \qquad s = j\omega \qquad \textbf{(14.10.1)}$$

But

$$R \parallel \frac{1}{sC} = \frac{R/sC}{R + 1/sC} = \frac{R}{1 + sRC}$$

Substituting this into Eq. (14.10.1) gives

$$\mathbf{H}(s) = \frac{R/(1 + sRC)}{sL + R/(1 + sRC)} = \frac{R}{s^2RLC + sL + R}, \qquad s = j\omega$$

or

$$\mathbf{H}(\omega) = \frac{R}{-\omega^2RLC + j\omega L + R} \tag{14.10.2}$$

Since $\mathbf{H}(0) = 1$ and $\mathbf{H}(\infty) = 0$, we conclude from Table 14.5 that the circuit in Fig. 14.39 is a second-order lowpass filter. The magnitude of $\mathbf{H}$ is

$$H = \frac{R}{\sqrt{(R - \omega^2RLC)^2 + \omega^2L^2}} \tag{14.10.3}$$

The corner frequency is the same as the half-power frequency, i.e., where $\mathbf{H}$ is reduced by a factor of $1/\sqrt{2}$. Since the dc value of $H(\omega)$ is 1, at the corner frequency, Eq. (14.10.3) becomes after squaring

$$H^2 = \frac{1}{2} = \frac{R^2}{(R - \omega_c^2RLC)^2 + \omega_c^2L^2}$$

or

$$2 = (1 - \omega_c^2LC)^2 + \left(\frac{\omega_cL}{R}\right)^2$$

Substituting the values of R, L, and C, we obtain

$$2 = (1 - \omega_c^2\, 4 \times 10^{-6})^2 + (\omega_c\, 10^{-3})^2$$

Assuming that ω_c is in krad/s,

$$2 = (1 - 4\omega_c^2)^2 + \omega_c^2 \qquad \text{or} \qquad 16\omega_c^4 - 7\omega_c^2 - 1 = 0$$

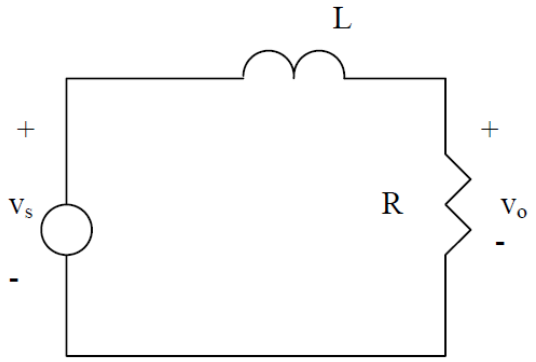
Solving the quadratic equation in ω_c^2, we get $\omega_c^2 = 0.5509$ and -0.1134. Since ω_c is real,

$$\omega_c = 0.742 \text{ krad/s} = 742 \text{ rad/s}$$

Practice Problem

Design an *RL* lowpass filter that uses a 40-mH coil and has a cutoff frequency of 5 kHz.

Solution:



The lowpass RL filter

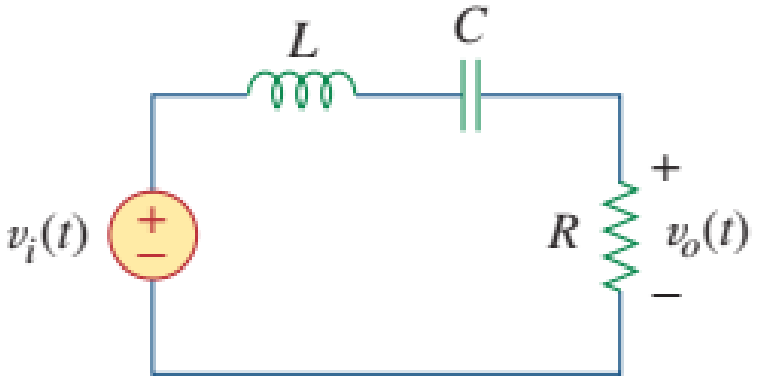
$$H = \frac{V_o}{V_s} = \frac{R}{R + j\omega L} = \frac{1}{1 + j\omega L / R}$$

$$\omega_c = \frac{R}{L} = 2\pi f_c$$

$$R = 2\pi f_c L = 2\pi x 5 x 10^3 \, x 40 x 10^{-3} = \underline{1.256 k\Omega}$$

Passive Filters: Bandpass Filter

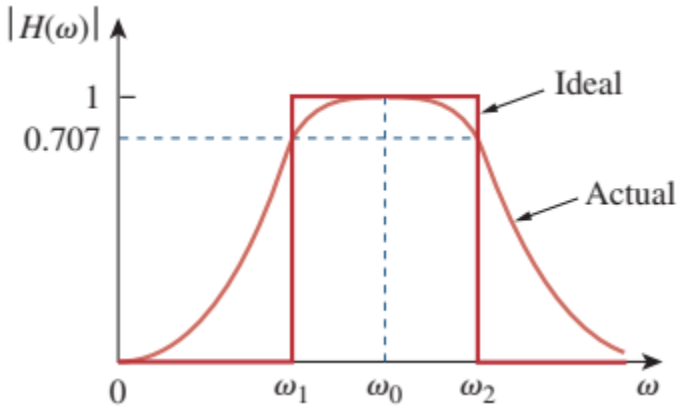
The *RLC* series resonant circuit provides a bandpass filter when the output is taken off the resistor



The transfer function is

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{R}{R + j(\omega L - 1/\omega C)}$$

We observe that $\mathbf{H}(0) = 0$, $\mathbf{H}(\infty) = 0$.

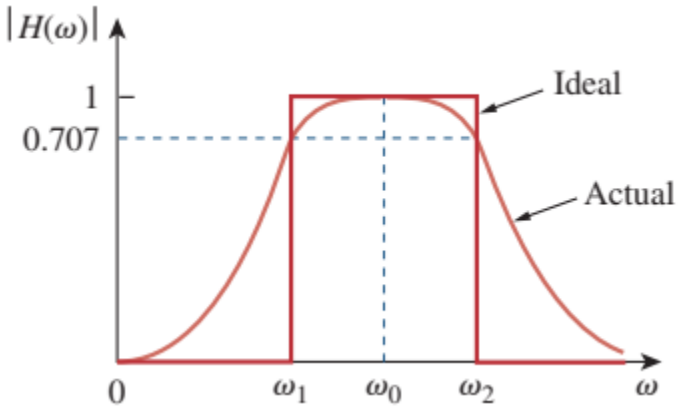


The bandpass filter passes a band of frequencies ($\omega_1 < \omega < \omega_2$) centered on ω_0, the center frequency, which is given by

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

A **bandpass filter** is designed to pass all frequencies within a band of frequencies, $\omega_1 < \omega < \omega_2$.

Bandpass Filter



ω_1 and ω_2 are called the *half-power frequencies.*

The half-power frequencies are obtained by setting input impedance Z equal to $\sqrt{2}R$,

$$\mathbf{Z} = \mathbf{H}(\omega) = \frac{\mathbf{V}_s}{\mathbf{I}} = R + j\omega L + \frac{1}{j\omega C}$$

$$\mathbf{Z} = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

$$\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} = \sqrt{2}R$$

Solving for ω, we obtain

$$\omega_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

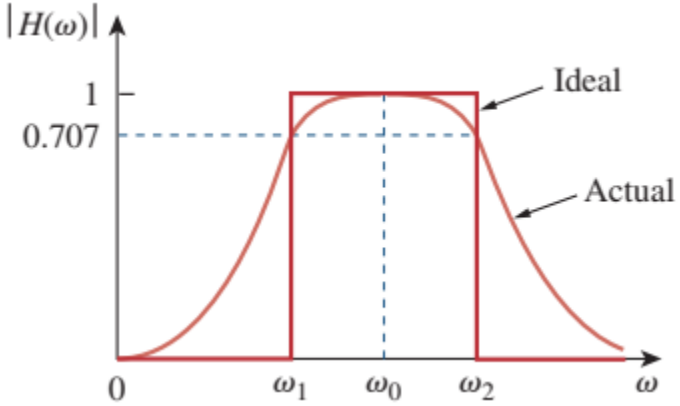
$$\omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

the resonant frequency. $\omega_0 = \sqrt{\omega_1 \omega_2}$

Bandpass Filter

the *bandwidth* B, which is defined as the difference between the two half-power frequencies,

$$B = \omega_2 - \omega_1$$



The "sharpness" of the resonance in a resonant circuit is measured quantitatively by the *quality factor* Q.

$$Q = 2\pi \frac{\text{Peak energy stored in the circuit}}{\text{Energy dissipated by the circuit in one period at resonance}}$$

$$Q = 2\pi \frac{\frac{1}{2}LI^2}{\frac{1}{2}I^2R(1/f_0)} = \frac{2\pi f_0 L}{R} \qquad Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 CR}$$

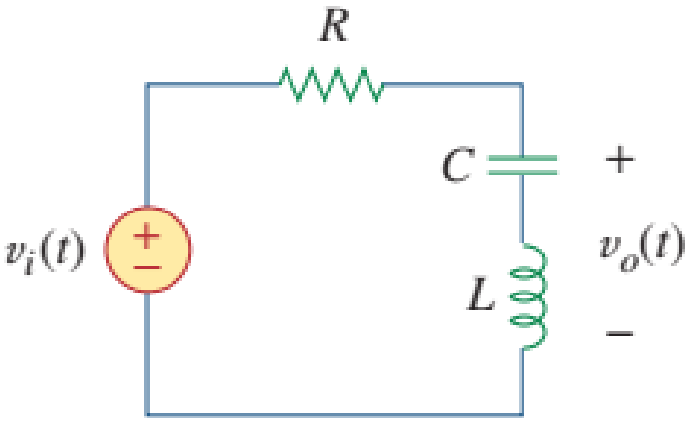
The relationship between the bandwidth B and the quality factor Q

$$B = \frac{R}{L} = \frac{\omega_0}{Q}$$

or $B = \omega_0^2 CR$.

Passive Filters: Bandstop Filter

A filter that prevents a band of frequencies between two designated values (ω_1 and ω_2) from passing is variably known as a *bandstop, band-reject*, or *notch* filter. A bandstop filter is formed when the output *RLC* series resonant circuit is taken off the *LC* series combination



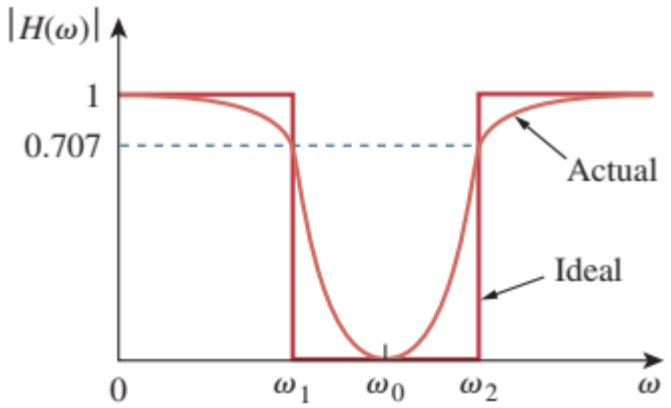
The transfer function is

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{j(\omega L - 1/\omega C)}{R + j(\omega L - 1/\omega C)}$$

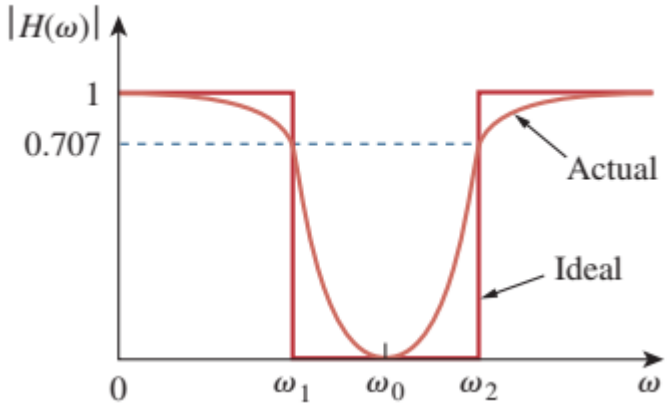
Notice that $\mathbf{H}(0) = 1$, $\mathbf{H}(\infty) = 1$.

Again, the center frequency is given by

$$\omega_0 = \frac{1}{\sqrt{LC}}$$



Bandstop Filter



ω_1 and ω_2 are called the *half-power frequencies.*

The half-power frequencies are obtained by setting input impedance Z equal to $\sqrt{2}R$,

$$\mathbf{Z} = \mathbf{H}(\omega) = \frac{\mathbf{V}_s}{\mathbf{I}} = R + j\omega L + \frac{1}{j\omega C}$$

$$\mathbf{Z} = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

$$\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} = \sqrt{2}R$$

Solving for ω, we obtain

$$\omega_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$\omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

the resonant frequency. $\omega_0 = \sqrt{\omega_1 \omega_2}$

the *bandwidth* B, which is defined as the difference between the two half-power frequencies,

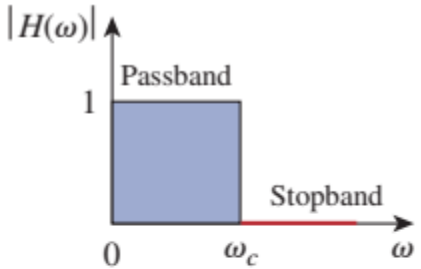
$$B = \omega_2 - \omega_1$$

A **bandstop filter** is designed to stop or eliminate all frequencies within a band of frequencies, $\omega_1 < \omega < \omega_2$.

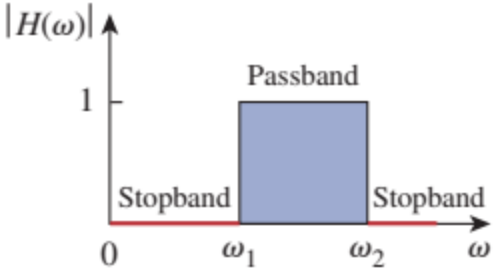
TABLE 14.5

Summary of the characteristics of ideal filters.

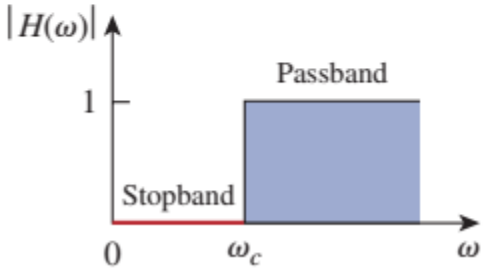
Type of Filter	$H(0)$	$H(\infty)$	$H(\omega_c)$ or $H(\omega_0)$
Lowpass	1	0	$1/\sqrt{2}$
Highpass	0	1	$1/\sqrt{2}$
Bandpass	0	0	1
Bandstop	1	1	0



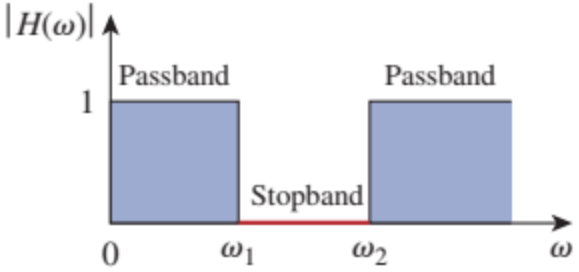
(a)



(c)



(b)



(d)

Example Problem 14.11

If the bandstop filter in Fig. 14.37 is to reject a 200-Hz sinusoid while passing other frequencies, calculate the values of L and C. Take $R = 150\ \Omega$ and the bandwidth as 100 Hz.

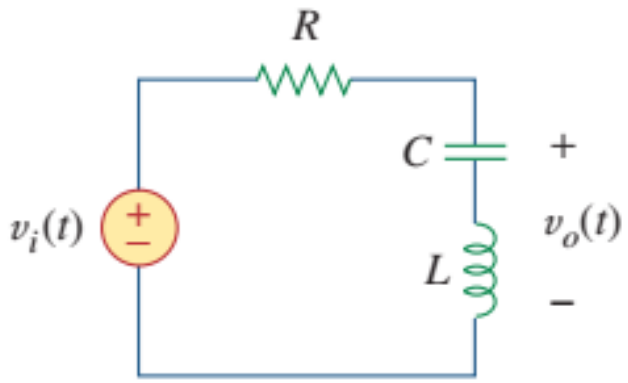
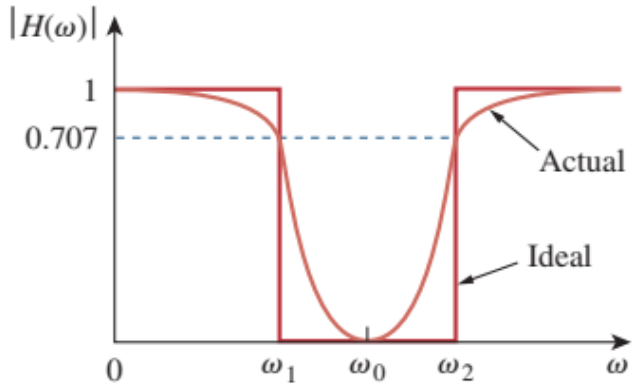


Figure 14.37
A bandstop filter.

Solution:

We use the formulas for a series resonant circuit in Section 14.5.

$$B = 2\pi(100) = 200\pi \text{ rad/s}$$



But

$$B = \frac{R}{L} \quad \Rightarrow \quad L = \frac{R}{B} = \frac{150}{200\pi} = 0.2387 \text{ H}$$

Rejection of the 200-Hz sinusoid means that f_0 is 200 Hz,

$$\omega_0 = 2\pi f_0 = 2\pi(200) = 400\pi$$

Since $\omega_0 = 1/\sqrt{LC}$,

$$C = \frac{1}{\omega_0^2 L} = \frac{1}{(400\pi)^2(0.2387)} = 2.653\ \mu\text{F}$$